

Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

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Abstract

001 Aggregate spatial accuracy can hide sharply differ-
002 ent capabilities within vision–language models (VLMs).
003 We study object-relative orientation as a diagnostic
004 case. On Visual Spatial Reasoning (VSR), moving from
005 a compact 2.2B model to a larger 7B VLM substan-
006 tially improves overall, depth, horizontal, and contain-
007 ment accuracy, but orientation changes only modestly.
008 Except for structured prompting, which degrades per-
009 formance, trained conditions on both models remain
010 in a narrow 62–66% orientation band. We test struc-
011 tured prompting, general and targeted LoRA, hard neg-
012 atives, visual/projector adaptation, frozen and object-
013 grounded representation probes, and object-centric de-
014 composition; none of the tested configurations reliably
015 closes the gap, and a single-annotator clean-label sen-
016 sitivity analysis shows that annotation ambiguity does
017 not explain it away.

018 Accuracy also fails to capture relational coherence.
019 LM-side adaptation significantly improves consistency
020 over zero-shot, while hard-negative training raises
021 facing/facing-away consistency from 66.0% to 77.7%
022 with facing accuracy unchanged at 68.9%; the addi-
023 tional pooled consistency gain over LM-only is not sig-
024 nificant. Cross-dataset evaluation on SITE shows that
025 VSR-trained adaptation does not transfer cleanly: it is
026 neutral overall and significantly harms official spatial-
027 relationship reasoning (75.1%→71.6%, $p = 0.004$). A
028 preregistered orientation-vocabulary slice is analyzed
029 as exploratory evidence rather than as direct replica-
030 tion of the VSR construct. The results point to a
031 persistent orientation bottleneck in which orientation
032 is not robustly decodable by the tested frozen-feature
033 readouts and relational decisions are not always propa-
034 gated coherently, while also showing that task-specific
035 spatial adaptation can transfer poorly.

1. Introduction

Spatial reasoning is central to multimodal systems that must interact with physical environments. A useful VLM should not only recognize objects, but determine whether one object is in front of another, whether it is facing another, whether two objects are parallel, and whether equivalent or complementary formulations yield logically compatible answers. These abilities matter for robotics, embodied agents, navigation, and scene understanding.

Existing work already shows that spatial relations are difficult for vision–language systems. VSR was designed as a controlled natural-image probe and reported particularly poor performance on object-orientation relations in earlier VLMs [5]. More recent controlled benchmarks likewise find that strong VLMs can fail on basic left/right and other grounded relations [4, 7], while broader evaluations continue to identify spatial orientation as a difficult factor [9]. Our question is therefore not whether spatial reasoning can fail, but whether the orientation weakness persists in modern generative VLMs of different sizes and after targeted intervention, and what the failure signature reveals about representation, reasoning, and transfer.

In our VSR experiments, the distinction between relation families is striking. Moving from SmolVLM2-2.2B-Instruct to the larger Qwen2-VL-7B-Instruct increases zero-shot overall accuracy from 74.0% to 80.9%. Horizontal reasoning rises from 70.2% to 84.8%, depth from 68.9% to 75.2%, and containment from 83.4% to 89.3%. Orientation, by contrast, moves only from 62.8% to 63.5%. We then evaluate interventions spanning prompting, language-side adaptation, targeted data construction, visual-side adaptation, representation probing, and explicit object-centric decomposition. Structured prompting significantly hurts performance; general and targeted LoRA improve aggregate accuracy but do not remove the orientation ceiling; hard negatives improve targeted relational coherence without a

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significant overall orientation-accuracy gain; and the tested visual-side LoRA configurations do not outperform the language-side control.

As a sensitivity analysis, removing examples flagged by a single-annotator audit increases orientation accuracy (65.7%→78.5% for 7B General LoRA and 66.4%→76.6% for hard-negative LoRA on the strictest 107-example subset) but does not eliminate error (2B structured shows no gain, 53.3% on both subsets). Because the removed examples were drawn from persistent failures, part of this gain is expected by construction; because this subset is not directly comparable to other relation families, we do not treat the absolute gap to containment or depth as a matched statistical comparison. The important result is that removing questionable examples raises scores but does not make orientation uniformly solved across conditions.

A second failure mode emerges from logical consistency. The zero-shot 7B model is consistent on only 36.9% of complementary facing/facing-away pairs. LM-side LoRA substantially improves coherence, and hard-negative training raises facing consistency to 77.7% while facing accuracy remains unchanged at 68.9%. The extra hard-negative gain is not significant in the pooled strict-family McNemar comparison, but the unchanged accuracy together with changed facing consistency demonstrates that the two metrics capture different behavior.

Finally, we test cross-dataset transfer. The 7B model is strong on SITE’s official spatial-relationship-reasoning category (75.1% raw; 59.2% chance-adjusted accuracy), yet the VSR-trained adapter does not generalize cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$). A preregistered, non-official orientation-vocabulary slice is much weaker in the aggregate (48.3% raw; 24.0% chance-adjusted) and shows no positive transfer, but—as we detail below—its composition limits what it can establish, and we therefore do not treat SITE as independent confirmation of the VSR orientation construct.

Our contributions are fourfold:

- Fine-grained diagnosis across two differently sized VLMs. Across ten VSR conditions and two differently sized models, several spatial families improve substantially while object-relative orientation remains unusually resistant.
- Robustness and mechanism analysis. A single-annotator clean-label sensitivity analysis, frozen/object-grounded probes, visual-side adaptation, and object-centric decomposition constrain purely annotation-based or single-component explanations.

- Accuracy-coherence separation. Complementary-relation tests show that relational consistency can change substantially while task accuracy remains fixed.
- Cross-dataset transfer analysis. SITE shows strong official spatial-relationship performance, while VSR-trained LoRA transfers poorly and significantly harms the official SITE category; the orientation-vocabulary slice is analyzed as exploratory evidence.

2. Related Work

Spatial reasoning benchmarks. VSR introduced a controlled natural-image benchmark of more than 10k image-text pairs spanning 66 spatial relations and explicitly highlighted orientation relations such as facing and parallel as difficult for the VLMs studied at the time [5]. Our work should therefore be read as a modern diagnostic extension rather than the first observation of an orientation weakness. What’sUp isolates basic spatial relations by holding object identities fixed while varying configurations, and found large gaps between humans and tested vision-language models [4]. GSR-Bench extends controlled grounded spatial evaluation across a larger set of multimodal models and model scales [7]. SITE broadens evaluation to single-image, multi-image, and video inputs and organizes tasks using cognitive-science dimensions including spatial visualization/orientation, intrinsic/extrinsic relations, and static/dynamic settings; its authors report spatial orientation as a persistent challenge [9]. Recent broad evaluations such as SpatialScore similarly argue that spatial intelligence remains fragmented across tasks and modalities [10].

From spatial perception to spatial reasoning. Recent work increasingly separates failures of perception, grounding, and reasoning rather than treating spatial ability as one scalar score. VSP decomposes visual spatial planning into finer perception and reasoning subtasks and reports deficiencies in both [11]. MM-Spatial focuses on 3D spatial understanding, including relationship prediction, metric estimation, and grounding, and shows that specialized spatial data and geometric inputs can improve multimodal models [1]. These studies motivate our relation-family and mechanism-level analysis. Our focus is narrower: object-relative orientation in natural images, with controlled tests of whether model size, language-side adaptation, visual-side adaptation, or explicit decomposition changes the bottleneck.

Adaptation and spatial specialization. We use LoRA, which adapts frozen models through trainable low-

rank updates [3], to compare general, targeted, hard-negative, projector, and vision-plus-projector interventions. This design lets us ask whether benchmark gains reflect general spatial competence or specialization. The external SITE transfer experiment is central here: VSR adaptation is neutral overall on SITE, does not improve the orientation subset, and significantly degrades SITE’s official spatial-relationship category. The result complements benchmark work by explicitly measuring negative as well as positive cross-dataset transfer.

Representation probing and logical consistency. Diagnostic probes can reveal whether information is accessible from frozen representations, but probe performance must be interpreted cautiously because probe capacity can itself learn the target task [2]. We therefore use linear and nonlinear probes, compare against majority baselines, add object-grounded features, and avoid claiming that weak decodability proves information is absent. Separately, recent work has begun to target spatial logical consistency directly. SAGE, for example, trains VLMs with geometric and linguistic duality operations to encourage coherent answers across transformed inputs [6]. Our consistency analysis is diagnostic rather than a new consistency-training framework: it shows that instance-level accuracy and complementary-relation coherence can diverge, and that hard negatives can change coherence without changing facing accuracy.

3. Experimental Setup

3.1. Benchmarks

VSR. We use VSR [5] as the primary diagnostic benchmark. The random-split test set contains 2,195 examples. The benchmark defines 64 relation labels; 61 appear in the evaluated test split. The orientation family contains facing, facing away from, parallel to, and perpendicular to, totaling 137 test examples. All main VSR results use the same held-out test set and deterministic True/False parsing.

SITE. We use SITE [9] for cross-dataset transfer evaluation only and do not train on SITE. The image evaluation contains 2,591 examples: 1,368 single-image and 1,223 multi-image examples. Our primary SITE subset is the official spatial-relationship-reasoning category ($n = 993$). We additionally report a transparently defined, non-official keyword-derived orientation subset ($n = 2,024$), analyzed as exploratory evidence rather than as direct replication of the VSR construct. SITE

results use the benchmark’s native multiple-choice format and chance-adjusted accuracy (CAA) in addition to raw accuracy.

3.2. Models and Conditions

Our compact model is SmolVLM2-2.2B-Instruct, and our larger model is Qwen2-VL-7B-Instruct [8]. We evaluate ten canonical VSR conditions: 2B zero-shot, 2B structured prompting, 2B General LoRA, 2B Targeted LoRA, 7B zero-shot, 7B General LoRA, 7B Targeted LoRA, 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B Vision+Projector LoRA. LoRA [3] is used for all parameter-efficient adaptation conditions.

General LoRA uses a proportional sample across relation families. Targeted LoRA allocates most training examples to weak orientation/depth/horizontal families with rehearsal from the remaining relations. Hard-negative LoRA introduces audited complementary orientation examples, primarily facing/facing-away inversions. Vision-side conditions adapt either the multimodal projector or upper vision blocks together with the projector. The frozen experimental tag paper-freeze-v1 records the final result set.

3.3. Metrics and Statistical Tests

For VSR we report accuracy overall and by spatial family. For SITE we report raw accuracy and CAA. Paired condition comparisons use McNemar tests when applicable. Clean-label orientation analysis reports the full orientation set and progressively stricter audited subsets. Logical consistency is evaluated by constructing complementary statements for strict relation pairs (left/right, front/behind, facing/facing-away) and testing whether the model assigns opposite verdicts as required. Parallel/perpendicular is treated separately as a soft complement because both relations can be false for oblique objects.

4. Main Results

4.1. Orientation Remains Weak Across Two Differently Sized VLMs

Table 1 shows the central pattern. Moving from the 2B model to the 7B model lifts zero-shot overall accuracy by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and containment by 5.9 pp, but orientation by only 0.7 pp. General 7B LoRA further raises overall accuracy to 84.7%, yet orientation reaches only 65.7%. Across trained conditions other than structured prompting, orientation remains in a narrow 62–66% band.

Structured prompting is a negative result: overall accuracy falls from 74.0% to 68.3% at 2B, while orientation falls from 62.8% to 53.3%. A paired exact

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4

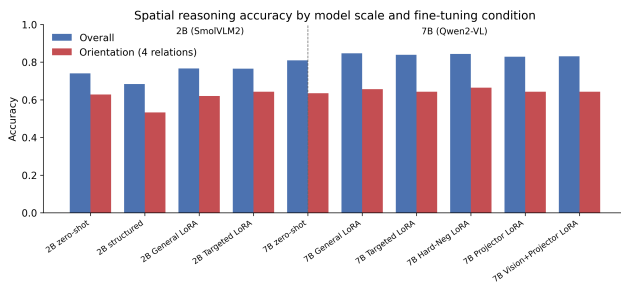


Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

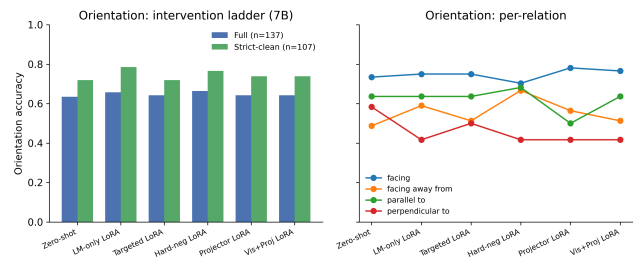


Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

McNemar test on the full test set confirms that the structured prompt is worse than the minimal baseline ($p \approx 2.2 \times 10^{-7}$). Explicit verbal decomposition therefore does not repair the bottleneck in the compact model.

4.2. Clean-Label Sensitivity Analysis

A single-annotator exploratory audit identified ambiguous or questionable orientation examples, so we repeat evaluation on progressively stricter subsets as a sensitivity analysis (full table in the supplementary). The best full-set orientation result is 66.4%; on the strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the evaluated sample, these values should not be read as a matched comparison against other relation families. The robust conclusion is narrower: removing flagged examples raises orientation scores but does not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise.

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations without fine-tuning the generative model. Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small ($n=34$) and the confidence interval overlaps the majority baseline. In short: orientation is not robustly decodable by the tested frozen-feature readouts.

To reduce the confound that global features might infer relations without identifying the relevant objects, we additionally extract subject/reference regions and construct grounded visual features from subject, object, difference, and elementwise-product representations. Object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [2], we

phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Detailed probe tables and plots are in the supplementary.

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, directly adapting visual or projector components might recover orientation performance. For the trained checkpoints evaluated here, it did not: projector LoRA and vision+projector LoRA both produced 64.2% orientation accuracy, below the 65.7% LM-only control, while overall accuracy was nominally lower (82.9% and 83.1% vs. 84.7%; unadjusted $p=0.0043/0.0122$; these effects do not survive a blanket multiple-comparison adjustment and are not load-bearing, App. D). This suggests that the tested low-rank adaptation of the visual pathway does not unlock the ceiling.

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline’s True/False verdicts agree with the end-to-end control on only 55.1–58.3% of the 127 boxed orientation statements; in the committed analysis (10 no-box statements scored wrong), verdicts diverge on 61–65 statements, and the control is correct while the pipeline is not in 44–47 of them. The committed pipeline scored agreement with the control rather than ground-truth accuracy, so we report the tested decomposition as failing to reproduce the generative model’s behavior, not as a direct accuracy comparison. Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not supply enough signal to close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The observed verdict patterns show what drives this: the model affirms either member of a facing pair only 17–20% of the time, a False-defaulting bias that yields both-False verdicts on

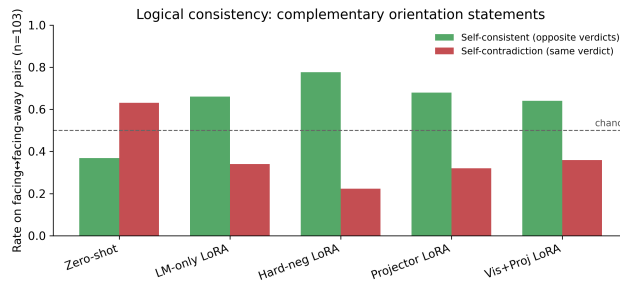


Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

63.1% of pairs and no both-True verdicts (0/103); expected consistency under verdict independence given these base rates is 30.2%, below the observed 36.9% (verdict table in the supplementary). The dominant phenomenon is a refusal to affirm either proposition, rather than a preference for contradictory affirmations.

LM-only LoRA raises facing consistency to 66.0% and front/behind consistency to 70.7%; pooled strict-family consistency is significantly better than zero-shot ($p < 0.0001$). Hard-negative training further raises facing consistency to 77.7%. Facing accuracy, however, is identical for LM-only and hard-negative LoRA at 68.9%. The additional hard-negative improvement is not significant in the pooled strict-family McNemar comparison ($p = 0.29$). Paired hard-negative supervision therefore provides a targeted intervention on complementary-relation coherence: it descriptively raises facing-pair consistency while leaving original facing accuracy unchanged, although the additional pooled consistency effect over LM-only is not significant. We do not claim a general coherence algorithm; the key result is separability: identical facing accuracy coexists with substantially different facing consistency. Training also changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% (LM-only) while both-True contradictions rise from 0% to 24.3%, a disclosed specialization trade-off.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE’s native multiple-choice protocol [9], with no SITE training. Because the preregistered decision rule resolved to branch 2 (the official category was not broadly weak), this adapter run is a post-protocol cross-dataset transfer check rather than a preregistered step; the protocol and its outcome are documented in the supplementary. The model is strong on the official spatial-relationship-reasoning cat-

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	-0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	-3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4

egory: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer on the independent benchmark; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The preregistered, non-official orientation-vocabulary slice is substantially weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% for the official category, and below the 54.2% / 31.1% all-images average) and shows no positive transfer under the adapter (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only 24% from the spatial-relationship category), we treat it as exploratory evidence rather than as direct replication of the VSR construct. A composition-confound analysis of the 2,591 zero-shot predictions (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and number of options, the orientation flag is not significantly associated with correctness (odds ratio 0.84, 95% CI [0.60, 1.16], $p = 0.29$), although it remains descriptively 7.5pp lower within the official spatial-relationship category itself (71.3% vs. 78.8%, $n=488/505$). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; $n=177$) yields 44.1% raw (95% CI [37.0, 51.4]; CAA -2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE’s official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in



Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation that orientation relations are difficult [5], while extending it to modern generative VLMs and a much broader intervention analysis.

Decodability and coherence both matter, as diagnostics. The probes show that orientation is not robustly decodable under the tested frozen-feature readouts, while consistency experiments expose large logical failures even for relations answered above chance. Neither result alone identifies a complete causal mechanism, and both are diagnostic rather than causal-localization claims. Together they support a mixed account in which orientation information is difficult to access reliably under the tested readouts and relational decisions are not always propagated coherently across complementary formulations.

Task adaptation can specialize rather than generalize. General LoRA improves VSR overall accuracy substantially, but its SITE transfer is neutral overall and significantly negative on the official spatial-reasoning subset. Hard negatives improve a targeted coherence behavior while producing relation-specific tradeoffs. Benchmark-specific fine-tuning gains should therefore not be equated automatically with general spatial-reasoning improvement.

8. Limitations

Our main diagnostic test contains only 137 orientation examples, and individual relation subsets such as perpendicular are smaller still ($n=12$); perpendicular results are therefore reported descriptively throughout. The single-annotator clean-label audit reduces ambiguity but also decreases sample size, and it is exploratory qualitative evidence rather than a ruling on label noise.

485 The SITE orientation slice is keyword-derived rather
486 than an official benchmark category; a composition-
487 confound analysis shows its aggregate score is largely
488 explained by task composition, so we do not treat SITE
489 as independent confirmation of the VSR orientation
490 bottleneck. Our probing methods test accessible decod-
491 ability under specific linear/MLP and pooling choices;
492 failure to decode does not prove information is absent.
493 Visual-side experiments use LoRA rather than full end-
494 to-end retraining. We study two differently sized VLMs
495 from different families; this is a cross-model compar-
496 ison, not a controlled scale experiment. All trained
497 conditions are single-checkpoint runs: the paired Mc-
498 Nemar tests quantify disagreement on the fixed test
499 examples, not training-run variability, and these single-
500 seed results do not estimate training-run variability.
501 The SITE evaluation is image-only; SITE video rea-
502 soning remains future work. Finally, several interven-
503 tions are evaluated as negative results, so lack of im-
504 provement should be interpreted for the tested config-
505 urations rather than as impossibility claims about all
506 prompting, adaptation, or architectural methods.

507 9. Conclusion

508 Aggregate spatial accuracy hides a persistent orien-
509 tation bottleneck in the VLMs studied here. A
510 larger VLM and standard adaptation substantially
511 improve several spatial families while leaving object-
512 relative orientation comparatively resistant. Auditing,
513 probing, visual-side adaptation, and decomposition
514 constrain simple explanations, while complementary-
515 relation tests show that accuracy and coherence are
516 distinct axes of spatial competence. Cross-dataset eval-
517 uation on SITE further shows that VSR-specific adap-
518 tation transfers poorly and can negatively transfer to
519 official spatial-relationship reasoning. The results mo-
520 tivate spatial evaluation that reports relation-specific
521 performance, transfer, and logical coherence rather
522 than relying on aggregate accuracy alone.

523 References

524 [1] Erik Daxberger, Nina Wenzel, David Griffiths, Haim-
525 ing Gang, Justin Lazarow, Gefen Kohavi, Kai Kang,
526 Marcin Eichner, Yinfei Yang, Afshin Dehghan, and Pe-
527 ter Grasch. MM-Spatial: Exploring 3d spatial under-
528 standing in multimodal LLMs. In Proceedings of the
529 IEEE/CVF International Conference on Computer Vi-
530 sion, pages 7395–7408, 2025. 2
531 [2] John Hewitt and Percy Liang. Designing and interpret-
532 ing probes with control tasks. In Proceedings of the
533 2019 Conference on Empirical Methods in Natural Lan-
534 guage Processing and the 9th International Joint Con-
535 ference on Natural Language Processing, pages 2733–

2743. Association for Computational Linguistics, 2019. 536
3, 4 537
[3] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan
Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and
Weizhu Chen. LoRA: Low-rank adaptation of large lan-
guage models. arXiv preprint arXiv:2106.09685, 2021. 538
3 539
[4] Amita Kamath, Jack Hessel, and Kai-Wei Chang. What’s “up” with vision-language models? investi-
gating their struggle with spatial reasoning. arXiv
preprint arXiv:2310.19785, 2023. 1, 2 540
541
[5] Fangyu Liu, Guy Emerson, and Nigel Collier. Visual
spatial reasoning. Transactions of the Association for
Computational Linguistics, 11:635–651, 2023. 1, 2, 3, 542
6 543
[6] Junming Liu, Yuqi Li, Yifei Sun, Maonan Wang, Piotr
Koniusz, Yirong Chen, and Ding Wang. Self-evolving
spatial reasoning in vision language models via geomet-
ric logic consistency. arXiv preprint arXiv:2605.18162,
2026. 3 544
545
[7] Navid Rajabi and Jana Kosecka. GSR-BENCH:
A benchmark for grounded spatial reasoning eval-
uation via multimodal LLMs. arXiv preprint
arXiv:2406.13246, 2024. 1, 2 546
547
[8] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhi-
hao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin
Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du,
Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou,
Jingren Zhou, and Junyang Lin. Qwen2-VL: Enhanc-
ing vision-language model’s perception of the world at
any resolution. arXiv preprint arXiv:2409.12191, 2024. 548
3 549
[9] Wenqi Wang, Reuben Tan, Pengyue Zhu, Jian-
wei Yang, Zhengyuan Yang, Lijuan Wang, Andrey
Kolobov, Jianfeng Gao, and Boqing Gong. SITE: to-
wards spatial intelligence thorough evaluation. In Pro-
ceedings of the IEEE/CVF International Conference
on Computer Vision, pages 9058–9069, 2025. 1, 2, 3, 5 550
551
[10] Haoning Wu, Xiao Huang, Yaohui Chen, Ya Zhang,
Yanfeng Wang, and Weidi Xie. SpatialScore: Towards
comprehensive evaluation for spatial intelligence. In
Proceedings of the IEEE/CVF Conference on Com-
puter Vision and Pattern Recognition, pages 31029–
31041, 2026. 2 552
553
[11] Qiucheng Wu, Handong Zhao, Michael Saxon, Trung
Bui, William Yang Wang, Yang Zhang, and Shiyu
Chang. VSP: Diagnosing the dual challenges of per-
ception and reasoning in spatial planning tasks for
MLLMs. In Proceedings of the IEEE/CVF Interna-
tional Conference on Computer Vision, pages 2270–
2280, 2025. 2 554
555
556
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558
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561
562
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